NLP Project 2: Preliminary Contribution Statement

COMS4054A / COMS7062A / COMS7066A • 14 October 2026

Group members: Sunay Master, Matthew Purkiss, Harshil Vallabh

Project: *Erased and penalised: 19th-century isiXhosa newspapers in modern NLP pipelines*

Research question. When the isiXhosa newspapers *Isigidimi samaXosa* (1870–1888) and *Imvo Zabantsundu* (1884–1894) from the isiXhosa Intellectual Traditions corpus (Schoots et al., 2023) are passed through a modern NLP pipeline, how much of the text is filtered out, and how much worse is it modelled? How much of that penalty comes from fixable spelling differences rather than genuine linguistic or cultural change?

Division of work

Each member contributes one third (33.3%) of the project.

Member	Share	Experiments	Writing and admin
Sunay	33.3%	A: pretraining filter audit	Introduction; background on data filtering; Experiment A sections; NeurIPS 2024 ethics checklist
Matthew	33.3%	B0: spelling normaliser and modern isiXhosa data; B1: tokeniser fertility	Dataset and orthography background; B0/B1 sections; Faculty of Science AI ethics statement
Harshil	33.3%	Data exploration and splits; B2: language-model experiment; combining the results	B2 section; limitations; Discussion; README.txt and requirements.txt

Sunay Master

- Run the filters used to build LLM training data (language ID, Gopher quality rules, C4 bad-words list, perplexity filter) on historical isiXhosa, historical English and modern isiXhosa, and measure how much of each survives.
- Find which filters remove the most text, and run them again on spelling-normalised text.
- Look at examples of removed text to see what kind of content is lost.

Matthew Purkiss

- Collect modern isiXhosa text (MasakhaNEWS, isiXhosa Wikipedia).
- Build a normaliser that converts historical spelling to modern spelling (e.g. *kuti* → *kuthi*), and check its accuracy on a hand-checked word list.
- Measure how much more modern tokenisers split up historical text than modern text.

Harshil Vallabh

- Corpus statistics, and train/validation/test splits by newspaper edition.
- Train character language models on raw vs normalised text, with and without added modern data, tuned on the validation set and evaluated once on the test set.

- Combine all results to show how much of the gap with modern text is due to spelling.

Shared by all members

- The two-page extended abstract (PDF), the full code, and checking every reference.
- The final contribution statement, which will report any changes to this plan and why.

Agreement

All group members have read and agree to this division of work.

Sunay Master	Matthew Purkiss	Harshil Vallabh
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____